AI-generated · Unofficial practice

Physics Mock

Answers and Marking Scheme

M: method mark A: answer mark C: communication / concept mark. For revision only; other valid methods should also be credited.

Section A Answers

1	2	3	4	5	6	7	8	9	10	11
C	C	D	B	A	A	D	B	A	A	D
12	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**
D	A	B	C	C	B	A	D	D	B	B
23	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**
D	C	C	A	C	B	A	B	C	D	B

1. C Heat lost by hot water = heat gained by cold water: $mc(70-30) = 0.40\,c(30-15)$, so $m = \dfrac{0.40 \times 15}{40} = 0.150\ \text{kg}$.

2. C Temperature is constant so average KE is unchanged: (1) correct. Molecules are pulled apart, so separation increases: (2) correct. The latent heat absorbed increases the molecular PE, so (3) is wrong.

3. D $n = \dfrac{2.0}{0.028} = 71.4\ \text{mol}$, $T = 290\ \text{K}$. $p = \dfrac{nRT}{V} = \dfrac{71.4 \times 8.31 \times 290}{0.050} = 3.44 \times 10^6\ \text{Pa}$.

4. B P and Q are $90°$ apart, so displacement $= \sqrt{40^2 + 40^2} = 56.6\ \text{m}$. Average velocity $= \dfrac{56.6}{30} = 1.89\ \text{m s}^{-1}$.

5. A The cylinder supports the plank from below at R; its reaction N is perpendicular to the plank, pointing up and to the left. Moments about P: $N \times 1.6 = 40 \times 1.0 \cos 30°$, $N = 21.7\ \text{N}$. Its horizontal component $N \sin 30° = 10.8\ \text{N}$ points left, so friction is $10.8\ \text{N}$ to the right.

6. A Treat X and Y as one system: $R + 10 \sin 30° = 7.0 \times 9.81$, so $R = 68.7 - 5.0 = 63.7\ \text{N}$.

7. D The change in momentum (impulse) is about the same, but the impact time is longer, so by $F = \dfrac{\Delta p}{\Delta t}$ the average force is smaller.

8. B Area (impulse) $= \frac{1}{2}(0.2)(4) + (0.4)(4) + \frac{1}{2}(0.2)(4) = 2.4\ \text{N s}$ (to the left). Taking right as positive: $0.50v = 0.50(3.0) - 2.4$, $v = -1.8\ \text{m s}^{-1}$, i.e. $1.8\ \text{m s}^{-1}$ to the left.

9. A On release the ball has both the tangential velocity of the platform and the velocity towards O, so the resultant points obliquely forward. With no horizontal force afterwards it moves in a straight line: path A.

10. A Two rope segments support the movable pulley, so pulling the free end $3.0\ \text{m}$ raises the load and pulley by $1.5\ \text{m}$. Gain in PE $= (400 + 40) \times 1.5 = 660\ \text{J}$. Energy against friction $= 800 - 660 = 140\ \text{J}$.

11. D Both drop the same height on smooth slides, so the final speeds are equal. Slide 1 is steep first, so its object gains speed earlier and has a larger average speed over the same length, arriving first.

12. D From the slopes: A before $2.0\ \text{m s}^{-1}$, after $-0.50\ \text{m s}^{-1}$; B after $1.5\ \text{m s}^{-1}$. $1.0(2.0) = 1.0(-0.50) + m(1.5)$, $m = 1.67\ \text{kg}$. KE before $2.0\ \text{J}$; after $0.125 + \frac{1}{2}(1.67)(1.5)^2 = 2.0\ \text{J}$, so the collision is elastic.

13. A After release the parcel keeps the same constant horizontal velocity as the drone, so it is always directly below the drone: (1) correct. The fall time $t = \sqrt{2h/g}$ is unchanged, so the horizontal displacement ut doubles: (2) correct. The landing speed $\sqrt{u^2 + 2gh}$ does not double: (3) wrong.

14. B Both points complete one turn in 24 h, so ω is the same: (1) correct. E rotates in a radius R, larger than $R\cos 22°$ for H, so $v = \omega r$ is larger: (2) correct. The circle of H is centred on the axis, so its centripetal acceleration points to the axis, not the centre of the Earth: (3) wrong.

15. C $\lambda = 8\ \text{cm}$. The two particles are $4\ \text{cm} = \frac{\lambda}{2}$ apart, in antiphase: (1) correct. The waveform shifts right by $(2 + 8n)\ \text{cm}$: $0.30 = (n + \frac{1}{4})T$, and $n = 1$ gives $T = 0.24\ \text{s}$: (2) correct. For a wave travelling right, the particle at $x = 0$ next takes the (negative) displacement of the particle just to its left, so it is moving downwards: (3) wrong.

16. C $v = \sqrt{T/\mu}$: the tension becomes 4 times, so v doubles. With f unchanged, $\lambda = \dfrac{v}{f}$ becomes 2λ.

17. B Frequency is fixed by the source. Waves are slower in shallow water, so $\lambda = v/f$ decreases and the waves bend towards the normal.

18. A $d\sin 20.0° = \lambda$, so $\sin\theta_2 = 2\sin 20.0° = 0.684$, $\theta_2 = 43.2°$. $n_{\max} < \dfrac{d}{\lambda} = \dfrac{1}{\sin 20.0°} = 2.92$, so $n = 0, \pm 1, \pm 2$: 5 fringes.

19. D Path difference $= 4.20 - 3.35 = 0.85\ \text{m}$. For antiphase sources, destructive interference needs path difference $= n\lambda$. At $800\ \text{Hz}$, $\lambda = 0.425\ \text{m}$ and the path difference is 2λ: destructive. At 200 Hz and 600 Hz it is 0.5λ and 1.5λ: constructive.

20. D Each reflection in a vertical plane mirror gives one lateral inversion; two reflections cancel, so the image appears to raise its right hand. Vertical mirrors do not invert top and bottom, so the image is upright.

21. B $u = 20$, $v = 60$: $\dfrac{1}{f} = \dfrac{1}{20} + \dfrac{1}{60}$, $f = 15\ \text{cm}$: (2) correct. Swapping u and v ($u = 60$, $v = 20$) also satisfies the lens formula with $u + v = 80$: (1) correct. The magnification is then $\dfrac{20}{60} = \dfrac{1}{3}$, a diminished image: (3) wrong.

22. B Ultrasound is sound above the 20 kHz limit of hearing: (1) correct. Sound travels faster in water than in air: (2) wrong. As a wave it can be reflected and diffracted: (3) correct.

23. D The electrostatic forces between the spheres form an action–reaction pair, internal to the system. Taking moments about O for the whole system (tensions give no moment): $2mgL\sin\theta_X = mgL\sin\theta_Y$, so $\sin\theta_Y = 2\sin\theta_X$ and $\theta_X < \theta_Y$, independent of the charges.

24. C The field of a negative charge points towards it. Near $x = -a$ the field points in $-x$ and is large and negative; near $x = +a$ it points in $+x$ and is large and positive; at the midpoint the fields cancel. So E rises steadily from large negative to large positive: graph C.

25. C $Q = It = 1.5 \times 40 \times 60 = 3600\ \text{C}$, $N = \dfrac{3600}{1.60 \times 10^{-19}} = 2.25 \times 10^{22}$.

26. A S open: $I = \dfrac{12}{2.0 + 4.0 + 2.0} = 1.5\ \text{A}$, $V = 1.5 \times 4.0 = 6.0\ \text{V}$. S closed shorts R_2: $I = \dfrac{12}{2.0 + 4.0} = 2.0\ \text{A}$, $V = 8.0\ \text{V}$.

27. C $F \propto \dfrac{I_1 I_2}{r}$. Q on R: opposite currents repel, $\dfrac{(2I)(3I)}{d} \to 6F_0$, away from Q (i.e. away from P). P on R: like currents attract, $\dfrac{(I)(3I)}{2d} \to 1.5F_0$, towards P. Resultant $= 6F_0 - 1.5F_0 = 4.5F_0$, away from P.

28. B The commutator keeps the external current one-way: (1) correct. $\varepsilon_{\max} = NBA\omega$, so doubling the speed doubles the peak, and the number of pulses per second also doubles: (2) correct. With the coil plane perpendicular to the field the flux is maximum but the sides move along the field lines without cutting them, so the e.m.f. is zero: (3) wrong.

29. A On the positive particle the electric force is down and the magnetic force is up, $qE = qvB$. (1): reversing B makes the magnetic force also downward, so the particle bends towards the lower, negative plate: correct. (2): at $2v$ the magnetic force doubles and wins, so it bends towards the upper, positive plate: correct. (3): $v = \dfrac{E}{B}$, so doubling E requires doubling B: wrong.

30. B $R = \dfrac{6.0^2}{3.0} = 12\ \Omega$. $V_{\text{rms}} = \dfrac{6.0}{\sqrt{2}}$, $P = \dfrac{V_{\text{rms}}^2}{R} = \dfrac{18}{12} = 1.5\ \text{W}$.

31. C $E = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{0.050 \times 10^{-9}} = 3.98 \times 10^{-15}\ \text{J} = 2.5 \times 10^{4}\ \text{eV} \approx 25\ \text{keV}$. X-rays are uncharged, so not deflected by a magnetic field; their wavelength is shorter than UV.

32. D Let the initial activity of P be x. 12 h is 3 half-lives of P and 1 of Q: $\dfrac{x}{8} + \dfrac{800 - x}{2} = 160$, $400 - \dfrac{3x}{8} = 160$, $x = 640\ \text{Bq}$.

33. B Conservation of nucleon number: $239 + 1 = 134 + A + 3$, so $A = 103$. (Charge: $94 = 54 + 40$, balanced.)

Section B Marking Scheme

1. (5 marks)

(a)(i) (2)

M1 $\theta = \dfrac{11.0 - 3.2}{18.4 - 3.2} \times 100$

A1 $= 51.3\ ^\circ\text{C}$

(a)(ii) (1)

C1 The length of the alcohol column varies linearly with temperature (uniform bore).

(b) (2)

M1 $\theta = \dfrac{11.0 - 3.2}{18.4 - 3.2} \times 98.0$

A1 $\theta = 50.3\ ^\circ\text{C}$; (a)(i) is an overestimate.

2. (6 marks)

(a) (2)

M1 $p_1V_1 = p_2V_2$: $1.5 \times 10^7 \times 0.012 = 1.2 \times 10^5 V_2$

A1 $V_2 = 1.50\ \text{m}^3$

(b)(i) (1)

A1 $1.50 - 0.012 = 1.49\ \text{m}^3$

(b)(ii) (3)

M1 $\text{Number} = \dfrac{1.488}{0.0080}$

A1 $= 186$ balloons

A1 $186 < 200$, so one cylinder is not enough.

3. (9 marks)

(a) (3)

C1 Gravitational PE is converted into KE (net loss of height 10.0 m).

M1 $\frac{1}{2}mv^2 = mg(12.0 - 2.0)$

A1 $v = 14.0\ \text{m s}^{-1}$

(b)(i) (2)

M1 $h = \dfrac{(14.0 \sin 35^\circ)^2}{2 \times 9.81} = 3.29\ \text{m}$

A1 Max height $= 3.29 + 2.0 = 5.29\ \text{m}$

(b)(ii) (3)

M1 $-2.0 = (14.0 \sin 35^\circ)t - \frac{1}{2}(9.81)t^2$

A1 $t = 1.86\ \text{s}$

A1 $x = (14.0 \cos 35^\circ)(1.86) = 21.3\ \text{m}$

(c) (1)

C1 Incorrect: landing speed depends only on the total drop of $12.0\ \text{m}$ (energy conservation), $\sqrt{2g(12.0)} = 15.3\ \text{m s}^{-1}$, independent of the angle.

4. (11 marks)

(a) (2)

A1 Weight vertically downward and normal reaction vertically upward.

A1 Tension towards the pulley and friction opposite to the motion.

(b)(i) (2)

A1 All points plotted correctly with labelled axes.

A1 Two straight lines meeting at $t = 1.0\ \text{s}$.

(b)(ii) (2)

A1 $a_1 = \dfrac{1.52}{1.0} = 1.52\ \text{m s}^{-2}$

A1 $a_2 = \dfrac{1.52 - 0.27}{1.0} = 1.25\ \text{m s}^{-2}$

(c) (2)

C1 At $t = 1.0\ \text{s}$ the weight hits the floor, the string becomes slack and the tension drops to zero.

C1 The only horizontal force is then friction, opposite to the motion, so the trolley decelerates.

(d) (3)

A1 $f = Ma_2 = 1.20 \times 1.25 = 1.50\ \text{N}$

M1 $mg - f = (M + m)a_1$

A1 $m = \dfrac{f + Ma_1}{g - a_1} = \dfrac{1.50 + 1.20(1.52)}{9.81 - 1.52} = 0.401\ \text{kg}$

5. (6 marks)

(6)

C1 Fix the box on the level floor of the bus with tape, with its long side along the direction of motion.

C1 While the bus accelerates steadily, find the water depths at the rear end h_r and front end h_f (read with the metre rule, or mark the water levels on the box / take a photo and measure afterwards), and the horizontal distance L between the two measuring points.

C1 Consider a small mass m of water at the surface: the net force from the surrounding water is perpendicular to the surface; $N\cos\theta = mg$.

C1 $N\sin\theta = ma$, hence $\tan\theta = \dfrac{a}{g}$.

C1 $\tan\theta = \dfrac{h_r - h_f}{L}$, so $a = \dfrac{g(h_r - h_f)}{L}$.

C1 Any one precaution: read when the water surface is steady; read at eye level / perpendicular to the scale; repeat on several runs and average; ensure the road is level.

6. (8 marks)

(a) (1)

C1 At each boundary $n_1 \sin\theta_1 = n_2 \sin\theta_2$; since the boundaries are parallel, the angle of refraction at one boundary equals the angle of incidence at the next, so $n\sin\theta$ is passed on unchanged.

(b)(i) (3)

M1 $1.00\sin 60^\circ = 1.47\sin\theta_p = 1.33\sin\theta_w$

A1 $\theta_p = 36.1^\circ$

A1 $\theta_w = 40.6^\circ$

(b)(ii) (2)

M1 In air below: $1.00\sin\theta = 1.00\sin 60^\circ = 0.866 < 1$

A1 It emerges, at 60° to the normal.

(c) (2)

M1 $1.52\sin 65^\circ = 1.38$, or critical angle $\sin c = \dfrac{1.33}{1.52}$, $c = 61.0^\circ$

A1 $1.38 > 1.33$ (or $65^\circ > 61.0^\circ$): total internal reflection, it cannot enter the water.

7. (10 marks)

(a) (2)

C1 The two slits act as coherent sources (diffracted waves from the same transmitter) and the waves superpose.

C1 Where the path difference is $n\lambda$ they interfere constructively (large reading); where it is $(n + \frac{1}{2})\lambda$, destructively (small reading).

(b) (2)

M1 Path difference $= 170.2 - 166.0 = 4.2\ \text{cm} = 1.5\lambda$

A1 Minimum; the second minimum below O.

(c) (3)

M1 $\Delta y = \dfrac{0.028 \times 1.20}{0.060}$

A1 $= 0.56\ \text{m}$

C1 The formula assumes small angles ($\lambda \ll a$); here $\sin\theta = \lambda/a = 0.47$, $\theta \approx 28°$, so $\sin\theta \neq \tan\theta$ and the formula underestimates.

(d)(i) (1)

A1 $n < \dfrac{a}{\lambda} = 2.14$, so $n = 0, \pm1, \pm2$: **5** maxima

(d)(ii) (2)

C1 The path difference can never exceed $a = 2.0\ \text{cm} < \lambda$, so $n\lambda$ with $n \geq 1$ is impossible.

A1 Only one maximum (at O).

8. (13 marks)

(a)(i) (2)

M1 $R = \dfrac{\rho l}{A} = \dfrac{1.7 \times 10^{-8} \times 1000}{\pi(1.25 \times 10^{-3})^2}$

A1 $= 3.46\ \Omega$

(a)(ii) (2)

M1 19 strands in parallel: $R = \dfrac{3.46}{19}$

A1 $= 0.182\ \Omega$

(a)(iii) (2)

M1 $R' = \dfrac{3.46}{17} = 0.204\ \Omega$

A1 Increase $= \dfrac{19}{17} - 1 = 11.8\%$

(b)(i) (2)

M1 $I = \dfrac{P}{V} = \dfrac{2.4 \times 10^6}{22 \times 10^3}$

A1 $= 109\ \text{A}$

(b)(ii) (2)

M1 $R = 16 \times 0.182 = 2.92\ \Omega$; $P = I^2R = 109^2 \times 2.92$

A1 $= 3.47 \times 10^4\ \text{W}$ (34.7 kW)

(b)(iii) (1)

A1 Current doubles, loss $\times 4$: $1.39 \times 10^5\ \text{W}$

(b)(iv) (2)

M1 Power received $= 2.40 - 0.139 = 2.26\ \text{MW}$

A1 $2.26 < 2.30$: cannot meet the needs.

9. (9 marks)

(a) (3)

C1 PQ carries current perpendicular to B, so it experiences a magnetic force (motor effect).

C1 By Fleming's left-hand rule, the force on PQ is vertically downward.

C1 This gives a moment tilting PQ down; the rider on the other side gives an opposite moment of equal size.

(b)(i) (2)

M1 $F \times 0.100 = 0.15 \times 10^{-3} \times 9.81 \times 0.080$

A1 $F = 1.18 \times 10^{-3}\ \text{N}$

(b)(ii) (2)

M1 $B = \dfrac{F}{IL} = \dfrac{1.18 \times 10^{-3}}{2.0 \times 0.050}$

A1 $= 1.18 \times 10^{-2}\ \text{T}$

(c)(i) (1)

C1 No. The force on PQ becomes upward, giving a moment in the same sense as the rider; the rider must be moved to the PQ side of the axis.

(c)(ii) (1)

C1 Any one: increase the current; use a lighter rider; use a longer PQ (or more turns on PQ); place PQ further from the axis.

10. (7 marks)

(a) (2)

M1 $\Delta m = 209.982874 - 205.974465 - 4.002603 = 0.005806\ \text{u}$

A1 $E = 0.005806 \times 931 = 5.41\ \text{MeV}$

(b)(i) (2)

M1 Momenta equal and opposite; $\text{KE} = \dfrac{p^2}{2m}$, so $\dfrac{\text{KE}_\alpha}{\text{KE}_{\text{Pb}}} = \dfrac{206}{4}$

A1 $\text{KE}_\alpha = \dfrac{206}{210} \times 5.41 = 5.30\ \text{MeV}$

(b)(ii) (1)

C1 Only two particles are produced; conservation of momentum (and energy) fixes how the released energy is shared, so the α particle always gets the same KE.

(b)(iii) (2)

M1 All the released energy ($5.41\ \text{MeV}$ per decay) becomes heat: $P = 4.0 \times 10^{11} \times 5.41 \times 1.60 \times 10^{-13}$

A1 $P \approx 0.35\ \text{W}$